ST 421
(Math Stat 1)

note: If $A \subseteq B \Rightarrow A \cap B = A$

Combinatorics - shortcuts developed for counting

$n! = n(n-1)(n-2)\cdots 2 \cdot 1$ → # of distinct ways the original n things can be shuffled

$C^n_r = \frac{P^n_r}{r!} = \frac{n!}{r!(n-r)!}$; since $r! \geq 1$, it follows that $C^n_r \leq P^n_r$; $C^n_r = C^n_{n-r}$

note: combinations are special cases of partitions - one where n things are split into two groups

Boole's inequality: $P(A \cup B) \leq P(A) + P(B) \Rightarrow P(\bigcup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} P(A_i)$

Ch. 1 - What is Statistics

Probabilist - reasons from known population to sample

Statistician - uses theory of probability to calculate probability of observed sample to infer from this the characteristics of an unknown population

*Probability is foundation of statistics

Ch. 2 - Probability

Ω - set of all elements under consideration (universal set)

Def: A is a subset of B if every point in A is also in B ($A \subset B$ proper or $A \subseteq B$ improper)

Def: The null (empty) set consists of no points ($\emptyset$); $\emptyset \subset$ {every set}

union: $A \cup B$ = {set of all points in A or B}; all points in at least one of sets

Intersection: $A \cap B$ = {set of all points in both A & B}; contains A & B simultaneously

"at least" ↓ my complement

complement: A^c = {all outcomes in Ω but not in A}; $A^c \cup A = \Omega$; $(A^c)^c = A$

Def: Two sets (A & B) are said to be disjoint (mutually exclusive) if $A \cap B = \emptyset$; in common no points

order of operations: commutativity & associativity hold

Distributive Laws: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$; $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

De Morgan's Laws: $(A \cap B)^c = A^c \cup B^c$; $(A \cup B)^c = A^c \cap B^c$

Experiment - process by which observation is made

Event - one/more outcomes from experiment

simple event - cannot be decomposed

compound event - can be decomposed into other events

Sample space (Ω) - set of all possible sample points

(Discrete) Ω - contains either a finite or countable number of distinct sample points

Finding Prob of Event for discrete Ω

(1) Sample-Point method

(2) Event-composition method

	order matters	order doesn't matter
w/ replacement	FTOC (mn rule) ex: email passwords	
w/o replacement	P^n_r ex: committee roles	C^n_r ex: committee (no) roles

$\bigcup_{i=1}^{\infty} A_i$ = {outcomes in at least one A_i}

$\bigcap_{i=1}^{\infty} A_i$ = {outcomes in all of A_i's}

events $\xrightarrow{P \text{ (function)}}$ numbers

Definition of Probability

(1) $P(A) \geq 0$ (2) $P(\Omega) = 1$

(3) If $A_i \cap A_j \neq 0$, $i \neq j$, then $P(A_1 \cup A_2 \cup A_3 \cup \cdots) = \sum_{i=1}^{\infty} P(A_i)$ - $P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$

$\Rightarrow P(A_1 \cup A_2 \cup \cdots \cup A_n) = \sum_{i=1}^{n} P(A_i)$

Theorem (mn rule; FTOC): with m elements $a_1, \ldots, a_m$ & n elements $b_1, \ldots, b_n$, it is possible to form $mn = m \times n$ pairs containing one element from each group

Def: An ordered arrangement of r distinct objects is called a permutation - P^n_r

$$\boxed{P^n_r = n(n-1)\cdots(n-r+1) = \frac{n!}{(n-r)!}}$$ · ordered combinations

Def: The number of combos of n objects taken r at a time - $C^n_r = \binom{n}{r}$

$$\boxed{\binom{n}{r} = C^n_r = \frac{P^n_r}{r!} = \frac{n!}{r!(n-r)!}}$$

· distinct combinations

$(P(B) > 0)$ conditional prob. $\boxed{P(A|B) = \frac{P(A \cap B)}{P(B)}}$

Theorem (multiplicative law)

$P(A \cap B) = P(A|B)P(B)$

(symmetry) $= P(B|A)P(A)$

If ind. $\Rightarrow P(A \cap B) = P(A)P(B)$

Independence: (i) $P(A|B) = P(A) \Leftrightarrow P(B|A) = P(B)$

(ii) $P(A \cap B) = P(A)P(B)$

If A + B are ind. ⇓ A + B^c are ind.

$P(A_1 \cap A_2 \cap A_3 \cap \cdots \cap A_k) = P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2) \cdots P(A_k|A_1 \cap A_2 \cap \cdots \cap A_{k-1})$

Theorem (Additive Law)

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

If ME: $P(A \cup B) = P(A) + P(B)$

$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap C) - P(A \cap B) - P(B \cap C) + P(A \cap B \cap C)$ (since $(A \cap B) \cap (A \cap C) = A \cap B \cap C$)

Theorem (Complementation Rule)

$P(A) = 1 - P(A^c)$

classical model: assume that all of the (finitely many) outcomes in Ω are equally likely, i.e.

$$\boxed{P(A) = \frac{\#A}{\#\Omega}}$$

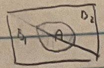
Def: For some positive integer k, let sets $B_1, \ldots, B_k$ be s.t. (1) $\Omega = B_1 \cup B_2 \cup \cdots \cup B_k$ (2) $B_i \cap B_j = \emptyset$ for $i \neq j$; then the collection $B_1, \ldots, B_k$ is a partition of Ω

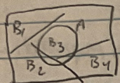
$\Rightarrow A = (A \cap B_1) \cup (A \cap B_2) \cup \cdots \cup (A \cap B_k)$

Q: what value is theory of probability (provides foundation/tools for statistical inference - objective of stats)

- provides theory/tools for calculating the probabilities of numerical events & hence prob. dist. for RV's discussed later
- numerical events of interest appear in a sample - & we wish to calculate the prob. of an observed sample to make an inference abt the target population

note: prob. dists we derive are models for the frequency dist's of populations of real data that occur (or would be generated) in nature; theoretical model for empirical dist. of data from population





$\boxed{\text{Theorem (Law of total Prob): Assume that } \{B_1, \ldots, B_k\} \text{ is a partition of } \Omega \text{ s.t. } P(B_i)>0 \text{ for } i=1,2,\ldots,k. \text{ Then } \boxed{P(A) = \sum_{i=1}^{k} P(A|B_i)P(B_i)}}$

- sometimes easier to calculate conditional prob than P(A) directly
- interested in second stage probability
- requires total prob. calculation; often interested in first stage prob.

Geometric series: $\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$ for all $r \in (-1,1)$

Theorem (Bayes Rule): Assume above conditions. Then $\boxed{P(B_j|A) = \frac{P(A \cap B_j)}{P(A)} = \frac{P(A|B_j)P(B_j)}{\sum_{i=1}^{k} P(A|B_i)P(B_i)}}$

p-series $\sum_{y=1}^{\infty} \frac{1}{y^p}$ — $p=1 \rightarrow$ diverges to ∞ — diverges for $p \leq 1$; converges for $p>1$

$\sum_{y=1}^{n} y = 1+2+\ldots+n = \frac{n(n+1)}{2}$

$\sum_{y=1}^{n} y^2 = 1^2+2^2+\ldots+n^2 = \frac{n(n+1)(2n+1)}{6}$

Theorem: Let X - RV w/ p(x). Then
(i) $0 \leq p(x) \leq 1$, x in $\mathcal{X}$
(ii) $\sum_x p(x) = 1$
(iii) $P(X \in B) = \sum_{x \in B} p(x)$

Def: A RV is a real-valued function for which the domain is a sample space
- said to be discrete if it can assume only a finite or countably inf. # of distinct values

Ch. 3 - Discrete RV's & their probability dists

probability dist. (discrete RV's): collection of probabilities; knowledge of prob. dist. for RV's associated w/ common types of experiments will eliminate need for solving same prob. problems

represented by formula, table, or graph for all y

Probability mass function: $p(y) = P(Y=y) = P\{\omega \in \Omega : Y(\omega) = y\}$ for all $y \in \mathcal{Y}$ - set of possible values that Y can take

$\boxed{\text{Theorem: For any discrete RV: (1) } 0 \leq p(y) \leq 1 \text{ for all } y \text{; (2) } \sum_y p(y) = 1 \text{ over nonzero prob. y-vals}}$

we wish to find numerical descriptive measures (parameters) for p(y) [prob. dist.]

$\boxed{E(Y) = \sum_y y\,p(y) = \mu}$ $\boxed{V(Y) = E\{(Y-\mu)^2\} = \sigma^2}$

$\boxed{E(c) = c, \text{ constant}}$ $\boxed{SD(Y) = \sqrt{V(Y)}}$

$\boxed{\text{Theorem: suppose } g(Y) \text{ is a real-valued function of } Y. \text{ Then } \boxed{E\{g(Y)\} = \sum_y g(y)p(y)} \text{ for all } y}$

$\boxed{E\{g_1(Y)+\ldots+g_k(Y)\} = E\{g_1(Y)\}+\ldots+E\{g_k(Y)\}}$ $\boxed{E\{c\,g(Y)\} = c\,E\{g(Y)\}}$

$\boxed{E(aY+b) = aE(Y)+b}$ $\boxed{V(aY+b) = a^2V(Y)}$, generalized result

$\boxed{V(Y) = E(Y^2) - [E(Y)]^2}$ ... shortcut formula

Binomial Expt
1. Fixed # of (identical) trials
2. Each outcome has S/F
3. Same S prob = p, $q = 1-p$
4. trials are independent
5. Y = # successes during n trials

binomial dist. w success prob. p:
$\boxed{p(y) = \binom{n}{y} p^y q^{n-y}}$
$y = 0,1,\ldots,n$ & $0 \leq p \leq 1$

Given $\underbrace{SFSSFSFS}_{n \text{ positions}} \rightarrow \underbrace{SSS\ldots SS}_{y}\ \underbrace{FFF\ldots FF}_{n-y} \rightarrow \underbrace{ppp\ldots pp}_{y}\ \underbrace{qqq\ldots qq}_{n-y} \rightarrow p^y q^{n-y}$ $\boxed{Y \sim Bin(n,p)}$

Satisfies necessary properties for p(y) since $\sum_y p(y) = \sum_{y=0}^{n} \binom{n}{y} p^y q^{n-y} = (q+p)^n = 1^n = 1.$

$\boxed{\mu = E(Y) = np}$ $\boxed{\sigma^2 = V(Y) = npq}$

Derivations illustrate common tricks: (i) $\sum_y p(y) = 1$ if p(y) is valid prob. fcn)
(ii) Find E(Y) by $E\{Y(Y-1)\}$

Geometric Prob. Dist

Y = # of trial on which first success occurs (possibly indefinite)

E_1: S
E_2: FS
⋮
E_k: $\underbrace{FF\ldots F}_{k-1}S$

$p(y) = P(Y=y) = P(E_y) = P(\underbrace{FFF\ldots F}_{y-1}S) = q^{y-1}p$

$\boxed{Y \sim Geo(p)}$

$\boxed{p(y) = q^{y-1}p}$, $y = 1,2,\ldots$, $0 \leq p \leq 1$

$\boxed{\mu = E(Y) = \frac{1}{p}}$

$\boxed{\sigma^2 = V(Y) = \frac{1-p}{p^2}}$

mean value inversely proportional to p

Some authors define Y^* = # failures before first success vs. Y = # trials on which first success occurs

note: $Y^* = Y - 1$
$\Rightarrow P(Y=y_0) = P(Y-1 = y_0-1) = P(Y^* = y_0-1)$.

$\boxed{P(Y>s) = (1-p)^s}$ for any positive integers s

Hypergeometric Prob. Dist: trials not independent & prob. dist. for Y cannot be approximated adequately by binomial RV (for when n is large relative to N - sampling w/o replacement)

Notation: N - pop size, Y - # successes in sample that contain (i) r (ii) N-r; sample of n units

$\boxed{p(y) = \frac{\binom{r}{y}\binom{N-r}{n-y}}{\binom{N}{n}}}$ $y = 0,1,\ldots,n$; $y \leq r$; $n-y \leq N-r$

notes: $\binom{a}{b} = 0$ if $b>a \Rightarrow p(y) \geq 0$

p(y) sums to 1 since $\sum_{i=0}^{n} \binom{r}{i}\binom{N-r}{n-i} = \binom{N}{n}$

$\boxed{Y \sim HypGeo(N,n,r)}$

$\boxed{\mu = E(Y) = \frac{nr}{N}}$

$\boxed{\sigma^2 = V(Y) = n\left(\frac{r}{N}\right)\left(\frac{N-r}{N}\right)\left(\frac{N-n}{N-1}\right)}$

* For fixed $p = \frac{r}{N}$, HypGeo prob. function converges to binomial function as N becomes large

Let $p = \frac{r}{N}$ & $q = 1-p = \frac{N-r}{N}$

$\Rightarrow \mu = np$, $\sigma^2 = npq\left(\frac{N-n}{N-1}\right) \Rightarrow \lim_{N \to \infty} \frac{N-n}{N-1} = 1$

$\Rightarrow \left(\frac{N-n}{N-1}\right)$ is adjustment appropriate when n is large relative to N $\Rightarrow \lim_{N \to \infty} \frac{\binom{r}{y}\binom{N-r}{n-y}}{\binom{N}{n}} = \binom{n}{y} p^y q^{n-y}$ for $p = \frac{r}{N}$

Ch. 3

Poisson Prob. Dist. $(Y \sim Pois(\lambda))$

- common model for count data
- involve info of form "counts per unit time"; if units don't match → make adjust. before applying formulas (*)

$\lambda = n\theta_n$, $\theta_n = \frac{\lambda}{n}$

$$\boxed{p(y) = \frac{\lambda^y}{y!} e^{-\lambda} \quad y = 0,1,\dots,\ \lambda > 0}$$

$\boxed{E(Y) = \lambda = V(Y)}$ Approximate/converge to binomial PDF for large n, small θ (*)

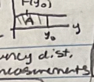
Ch. 4 - Continuous RV's & Their Prob. Dist.

Prob. Dist $\boxed{\text{CDF: } F(y) = P(Y \le y)}$, $-\infty \le y \le \infty$

continuous CDF ⇒ continuous RV

$\boxed{\text{PDF: } f(y) = F'(y) = \frac{d}{dx} F(y)}$

$\boxed{F(y) = \int_{-\infty}^{y} f(t)\,dt}$ where t - dummy var



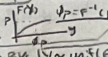
PDF - theoretical model for frequency dist. (hist) of population of measurements

(i) $f(y) \ge 0$
(ii) $\int_{-\infty}^{\infty} f(y)\,dy = 1$

$\boxed{P(Y \in A) = \int_A f(y)\,dy}$

Summaries $\boxed{F(\phi_p) = P(Y \le \phi_p) = p}$ $\boxed{F(y) \overset{set}{=} p}$ → solve for y to find ϕ_p

$\boxed{E(Y) = \int_{-\infty}^{\infty} y f(y)\,dy}$ p^{th} quantile, $100p^{th}$ percentile



$\phi_p = F^{-1}(p)$

$\boxed{E\{g(y)\} = \int_{-\infty}^{\infty} g(y) f(y)\,dy}$

$\boxed{E(c) = c}$ $\boxed{E\{cg(Y)\} = cE\{g(Y)\}}$

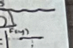
$\boxed{E\{g_1(Y) + \dots g_k(Y)\} = E\{g_1(Y)\} + \dots E\{g_k(Y)\}}$

$\boxed{E\{(Y-\mu)^2\} = V(Y) = E(Y^2) - E^2(Y)}$

special continuous RV's

Uniform RV's $\boxed{Y \sim unif(\theta_1, \theta_2)}$

$\boxed{f(y) = \begin{cases} 1/\theta_2 - \theta_1, & \theta_1 \le y \le \theta_2 \\ 0, & \text{otherwise} \end{cases}}$



$\boxed{F(y) = \begin{cases} \frac{y - \theta_1}{\theta_2 - \theta_1}, & \theta_1 \le y \le \theta_2 \end{cases}}$ width height

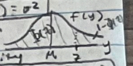
unif important theoretically $\boxed{E(Y) = \frac{\theta_1 + \theta_2}{2}}$ $\boxed{V(Y) = \frac{(\theta_2 - \theta_1)^2}{12}}$

Normal RV's $\boxed{Y \sim N(\mu, \sigma^2)}$

$\boxed{E(Y) = \mu}$ $\boxed{V(Y) = \sigma^2}$

$\boxed{f(y) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{\frac{-(y-\mu)^2}{2\sigma^2}\right\}, -\infty \le y \le \infty}$

no closed-form expression for F



$\boxed{Z = \frac{X - \mu}{\sigma} \sim N(0,1)}$ standardizing preserves normality

use standard normal tbl, observe direction of probs (tail)

$\boxed{\Phi(z) = P(Z \le z)}$ $\boxed{P(Z > z) = 1 - \Phi(z)}$

Normal Quantiles
(1) Find ϕ_p for Z - denote Z_{ϕ_p} → z-table
(2) Get ϕ_p for Y ⇒ $\boxed{\phi_p = \mu + Z_{\phi_p}\sigma}$

Gamma (class) RV's (continuous) wait times until an event occurs

α-shape β-scale

$\boxed{Y \sim Gamma(\alpha, \beta)}$ $\boxed{f(y) = \begin{cases} cy^{\alpha-1} e^{-y/\beta}, & y \ge 0 \\ 0, & \text{otherwise} \end{cases}}$ $c = \frac{1}{\beta^\alpha \Gamma(\alpha)}$, $\Gamma(\alpha)$ - gamma fx

→ no CDF expression

$\boxed{E(Y) = \alpha\beta}$ $\boxed{V(Y) = \alpha\beta^2}$ 2 special cases: (i) chi-square (χ^2) (ii) Exponential (EXP)

Chi-Square RV w/ r df i.f.f. $\boxed{Y \sim \chi^2(r)}$ $\alpha = \frac{r}{2}$, $\beta = 2$

if $Z \sim N(0,1)$, then $Z^2 \sim \chi^2$ w/ $r = 1$ df $\boxed{E(Y) = r}$ $\boxed{V(Y) = 2r}$

χ^2 table w/ probabilities exist

Exponential RV $\boxed{Y \sim EXP(\beta)}$ Gamma s.t. $\alpha = 1$

$\boxed{f(y) = \begin{cases} \frac{1}{\beta} e^{-y/\beta}, & 0 \le y < \infty \\ 0, & \text{otherwise} \end{cases}}$ $\boxed{F(y) = \begin{cases} 0, & y \le 0 \\ 1 - e^{-y/\beta}, & y > 0 \end{cases}}$

$\boxed{E(Y) = \beta}$ $\boxed{V(Y) = \beta^2}$

Beta (class) RV's α, β are shape 1, shape 2

$\boxed{Y \sim Beta(\alpha, \beta)}$ $\boxed{f(y) = \begin{cases} cy^{\alpha-1}(1-y)^{\beta-1}, & 0 \le y \le 1 \\ 0, & \text{otherwise} \end{cases}}$

note: $\alpha, \beta > 0$
→ no CDF expression

$\boxed{E(Y) = \frac{\alpha}{\alpha + \beta}}$ $\boxed{V(Y) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}}$

Special cases (i) $\alpha = \beta = 1$ ⇒ $Y \sim unif(0,1)$
(ii) $(\alpha = 1) \cup (\beta = 1)$ ⇒ $f(y)$ simplifies

Ch. 5 - Multivariate Prob. Dist.'s

$\boxed{F(y_1, y_2) = P(Y_1 \le y_1, Y_2 \le y_2)\ {-\infty < y_1 < \infty},\ {-\infty < y_2 < \infty}}$

$\boxed{p(y_1, y_2) = P(Y_1 = y_1, Y_2 = y_2),\ -\infty < y_1 < \infty,\ -\infty < y_2 < \infty}$

$\boxed{F(y_1, y_2) = \int_{-\infty}^{y_1}\int_{-\infty}^{y_2} f(t_1, t_2)\,dt_2\,dt_1}$ → use volume trick if PDF is constant

$\boxed{(\cdot)\ p(y_1, y_2) \ge 0\ \forall\ y_1, y_2}$ $\boxed{(\cdot) \int\int f(y_1, y_2)\,dy_1\,dy_2 = 1}$ → continuous CDF ⇒ continuous (Y_1, Y_2)

$\boxed{(\cdot) \sum_{y_1, y_2} p(y_1, y_2) = 1}$ $\boxed{P\{(Y_1, Y_2) \in A\} = \iint_A f(y_1, y_2)\,dy_1\,dy_2}$

$\boxed{P\{(Y_1, Y_2) \in A\} = \sum_{y_1, y_2 \in A} p(y_1, y_2)}$ sketch appropriate region of integration first to set up correct integral

most often construct table to facilitate calculations

Marginal & Conditional Prob. Dist.

$\boxed{p_1(y_1) = \sum_{\text{all } y_2} p(y_1, y_2)}$ $\boxed{f_1(y_1) = \int_{\text{all } y_2} f(y_1, y_2)\,dy_2}$

marginal PMF/PDF derived from joint PMF/PDF

$\boxed{p(y_1 | y_2) = \frac{p(y_1, y_2)}{p_2(y_2)},\ p_2(y_2) > 0}$

→ look at the effect on X_1 of knowing vs. not knowing X_2

$\boxed{f(y_1 | y_2) = \frac{f(y_1, y_2)}{f_2(y_2)}}$

Independence of RV's $\boxed{F(y_1, y_2) = F_1(y_1) F_2(y_2)}$

$\boxed{f(y_1, y_2) = f_1(y_1) f_2(y_2)\ \forall (y_1, y_2) \text{ pairs}}$

$\boxed{p(y_1, y_2) = p_1(y_1) p_2(y_2)\ \forall (y_1, y_2) \text{ pairs}}$

more intuitively

$\boxed{f(y_1 | y_2) = f_1(y_1)}$

$\boxed{p(y_1 | y_2) = p_1(y_1)}$

Multiplicative Rule: $\boxed{p(y_1, y_2) = p(y_1 | y_2)\, p_2(y_2)}$

recall symmetry $\boxed{f(y_1, y_2) = f(y_1 | y_2) f_2(y_2)}$

Shortcut: Y_1, Y_2 cannot be ind. unless range of (Y_1, Y_2) shaped like rectangle

2 types of Q:
(1) From joint dist., show that the RV's are ind.
(2) Assume (Y_1, Y_2) ind., and use marginal dist. to construct joint dist. to do calculations

Expected values of RV Functions

$\boxed{E\{cg(Y_1, Y_2)\} = cE\{g(Y_1, Y_2)\}}$

$\boxed{E(c) = c}$ $\boxed{E\{g_1(Y_1, Y_2) + \dots + g_k(Y_1, Y_2)\} = E\{g_1(Y_1, Y_2)\} + \dots + E\{g_k(Y_1, Y_2)\}}$

If ind.
$\boxed{E\{g(Y_1)h(Y_2)\} = E\{g(Y_1)\}E\{h(Y_2)\}}$

$\boxed{E\{g(Y_1, Y_2)\} = \sum_{\text{all } y_1}\sum_{\text{all } y_2} g(y_1, y_2) p(y_1, y_2)}$ $\boxed{E\{g(Y_1, Y_2)\} = \int_{\text{all } y_1}\int_{\text{all } y_2} g(y_1, y_2) f(y_1, y_2)\,dy_2\,dy_1}$

$\boxed{E(Y_1) = \int y_1 f_1(y_1)\,dy_1}$ $\boxed{V(Y_1) = \int (y_1 - \mu_1)^2 f_1(y_1)\,dy_1}$

Covariance of Two RV's

think of dependence of two RV's Y_1, Y_2 ⇒ one var - e.g. Y_1 - increases/decreases as Y_2 changes

$\boxed{COV(Y_1, Y_2) = E\{(Y_1 - \mu_1)(Y_2 - \mu_2)\} = E(Y_1 Y_2) - E(Y_1)E(Y_2)}$

→ magnitude indicates how strongly (Y_1, Y_2) are related → cov has scale
→ no scale → $\boxed{\rho(Y_1, Y_2) = \frac{COV(Y_1, Y_2)}{\sqrt{V(Y_1)V(Y_2)}} = \frac{C(Y_1, Y_2)}{\sigma_1 \sigma_2}}$ $\boxed{-1 \le \rho(Y_1, Y_2) \le 1}$
→ describes strength of linear relationship

$\boxed{\text{Independence} \Rightarrow \text{Zero covariance (correlation)} \quad \not\Leftarrow \text{ (converse not true)}}$

Summaries of Linear Functions of RV's

$\boxed{E\{a_1 Y_1 + a_2 Y_2\} = a_1 E(Y_1) + a_2 E(Y_2)}$

$\boxed{V\{a_1 Y_1 + a_2 Y_2\} = a_1^2 V(Y_1) + a_2^2 V(Y_2) + 2a_1 a_2 C(Y_1, Y_2)}$

note: $C(Y_1, Y_2) = 0$ if Y_1, Y_2 are ind.

$\boxed{E\{(\sum_{i=1}^{n} a_i Y_i) = \sum_{i=1}^{n} a_i E(Y_i)}$

$\boxed{V(\sum_{i=1}^{n} a_i Y_i) = \sum_{i=1}^{n} a_i^2 V(Y_i) + 2\sum_{i<j} a_i a_j C(Y_i, Y_j)}$

Let $\bar{Y}_n = \frac{1}{n}\sum_{i=1}^{n} Y_i$ (mean from $Y_1, \dots, Y_n$)

"Fundamental theorem of stats" $\boxed{E(\bar{Y}_n) = \mu}$ $\boxed{V(\bar{Y}_n) = \frac{\sigma^2}{n}}$

Let $Y_1, \dots, Y_n$ be ind. w/ common mean μ and variance σ^2 - both finite → $\bar{Y}_n$ unbiased
→ variance decreasing function of n, i.e. n^{-1}

Probability Inequalities

situations for no CDF (no closed form solution)
Find approximate probabilities (no analytical solution)

Markov: Let Y be positive RV. Then $\boxed{P(X > a) \le \frac{E(X)}{a}}$, $a > 0$

Chebysev: Let Y be RV s.t. $\mu = E(X)$, $\sigma^2 = V(X)$.

$$\boxed{P(|X - \mu| > a) \le \frac{\sigma^2}{a^2}}$$

Tchebysheff's: determine lower bound for the prob. that the RV Y of interest falls w/in interval $\mu \pm k\sigma$

Let Y be RV s.t. $k > 0$ w/ μ, σ^2: $\boxed{P(|Y - \mu| < k\sigma) \ge 1 - \frac{1}{k^2}}$ ⟺ $\boxed{P(|Y - \mu| \ge k\sigma) \le \frac{1}{k^2}}$

Ch. 6 - Functions of RV's

If $Y_1, \dots, Y_n$ treated as RV's - then a statistic is also a RV
→ want to derive dist.'s of statistics (e.g. $\bar{Y}_n$) analytically

Method of Dist. Functions

Start w/ RV Y, interest in $X = h(Y)$

Goal: get PDF, etc. from PDF of Y

recall: $f_X(x) = \frac{d}{dx} F_X(x)$

Strategy: Start w/ CDF of X wish to calculate probabilities, expected values, etc. from PDF of X
→ reasonable strategies for such calculations proceed in ch. 6

NOTES:
- Normal linear transformations are normal
- Square of a standard normal is a χ^2 RV (special type of Gamma)

note: find range of X before finding inverse

Method of Transformation

one-to-one - function if its strictly increasing or decreasing (monotonic) on its domain, e.g.
(·) linear functions, e.g. $ay + b$
(·) (some) nonlinear fx's, e.g. $h(y) = e^y$
(·) $h = y^2$ is not one-to-one
(·) trig are not

one-to-one functions are invertible, denoted by h^{-1}
→ setting $x = h(y)$ and solving for y in terms of x, e.g. $x = -\log y \Leftrightarrow y = e^{-x}$ ⇒ $h^{-1}(x) = e^{-x}$

- Basic strategy (start w/ CDF → integrate to find PDF) always works
→ able to apply shortcut for only 1-to-1 transformations

Transformation Rule: suppose $X = h(Y)$ where h is 1-to-1 w/ inverse h^{-1}.

$$\boxed{f_X(x) = f_Y(h^{-1}(x)) \left|\frac{dh^{-1}(x)}{dx}\right|}$$

change of vars (discrete)

$\sum_{x_2-x_1}^{\infty} \left(\frac{1}{2}\right)^{x_2-x_1}$ let $t = x_2 - x_1$

$\sum_{t=0}^{\infty} \frac{1}{2}^{t} = \frac{1}{1-\frac{1}{2}} = 2$ ■

IBP $\int u\,dv = uv - \int v\,du$ LIATE

$\int_0^{\infty} \underbrace{y}_{u} \underbrace{e^{-y}dy}_{dv}$ $u = y$, $\int dv = \int e^{-y}dy = -e^{-y} = v$

$= y(-e^{-y})\Big|_0^{\infty} - \int_0^{\infty} (-e^{-y})\,dy$

Probability integral transform (pseudo-RV's w/ unif(0,1) RV's)

Let $U \sim unif(0,1)$ and take any CDF F that has inverse F^{-1}. Then

$X = F^{-1}(U)$ has CDF F, i.e. $P(X \leq x) = F(x)$

→ If we can produce $U \sim unif(0,1)$ using computer, we can transform this to a RV X that has whatever (continuous) dist. we like — w/ CDF F — provided that we can evaluate the CDF F^{-1}

For **discrete cases**: deal directly w/ PMF of starting RV (no trick/shortcut) → simulations vital in modern statistics

Expectations: let $g(x_1, x_2) = x_1 + x_2$

$E\{(X_1+X_2)\} = \iint (x_1+x_2) f(x_1, x_2)\,dx_2\,dx_1$

$= \iint [x_1 f(x_1,x_2) + x_2 f(x_1,x_2)]\,dx_2\,dx_1$

$= \iint x_1 f(x_1,x_2)\,dx_2\,dx_1 + \iint x_2 f(x_1,x_2)\,dx_2\,dx_1$

$= \int x_1 f_1(x_1)\,dx_1 + \int x_2 f_2(x_2)\,dx_2$

$\left(f_1(x_1) = \int_{\text{all } x_2} f(x_1,x_2)\,dx_2 \Rightarrow f_2(x_2) = \int_{\text{all } x_1} f(x_1,x_2)\,dx_1\right)$

Multivariate (Bivariate) transformations similar techniques available for making transformations from a bivariate RV (Y_1, Y_2) to another bivariate RV $(X_1, X_2) = h(Y_1, Y_2)$ → only focus on 1-1 such transformation

bivariate transformation rule:

$$\boxed{f_{X_1,X_2}(x_1,x_2) = f_{Y_1,Y_2}(h_1^{-1}(x_1,x_2), h_2^{-1}(x_1,x_2)) \left| \det \begin{pmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} \\ \frac{\partial y_2}{\partial x_1} & \frac{\partial y_2}{\partial x_2} \end{pmatrix} \right|}$$

→ Jacobian

Expectations: $f(x_1,x_2) = 4x_1x_2$ $\quad f_1(x_1) = 2x_1$, $f_2(x_2) = 2x_2$

$E(X_1X_2) = \int_0^1\int_0^1 x_1x_2 \cdot 4x_1x_2\,dx_1\,dx_2$

$= 4\int_0^1\int_0^1 x_1^2x_2^2\,dx_1\,dx_2$

$= 4\left(\int_0^1 x_2^2\,dx_2\right)\left(\int_0^1 x_1^2\,dx_1\right)$

$= \left(\int_0^1 x_2 \cdot 2x_2\,dx_2\right)\left(\int_0^1 x_1 \cdot 2x_1\,dx_1\right)$

$= \quad E(X_2) \quad E(X_1)$

$\rightarrow E(X_1X_2) = E(X_1)E(X_2)$

Fact: the sum of ind. exponential RV's is a gamma RV — bookkeeping for change-of-variables

Simulating (bivariate) Normal RV's | **note**: find range of (X_1, X_2) before finding inverse

Let $U_1, U_2 \sim Unif(0,1)$

Then $X_1, X_2 \sim N(0,1)$ if

$X_1 = \sqrt{-2\log U_1}\cos(2\pi U_2)$

$X_2 = \sqrt{-2\log U_1}\sin(2\pi U_2)$

Moments & Moment-Generating Functions

- set of numerical measures (under certain conditions) uniquely determine $p(y)$

k^{th} moment (of a RV Y) taken abt origin is $\boxed{E(Y^k)}$

denoted **μ'_k** (·) $E(Y) = \mu'_1 = \mu$

(·) $E(Y^2) = \mu'_2$ employed for $V(Y) = E(Y^2) - E^2(Y)$

k^{th} central moment (of Y): k^{th} moment of RV Y taken abt its mean $\boxed{E\{(Y-\mu)^k\}}$ — **μ_k**

(·) $\sigma^2 = V(Y) = E\{(Y-\mu)^2\} = \mu_2$

Fundamental theorem of stats

$E(\bar{Y}_n) = E\left(\frac{1}{n}\sum_{i=1}^{n} Y_i\right) = \frac{1}{n}E\left(\sum_{i=1}^{n} Y_i\right) = \frac{1}{n}(n\mu) = \mu$.

$V(\bar{Y}_n) = V\left(\frac{1}{n}\sum_{i=1}^{n} Y_i\right) = \left(\frac{1}{n}\right)^2 V\left(\sum_{i=1}^{n} Y_i\right) = \frac{1}{n^2}(n\sigma^2) = \frac{\sigma^2}{n}$.

Moment Generating Functions

$\boxed{m_X(t) = E(e^{tX}) = \int e^{tx} f_X(x)\,dx}$

$\boxed{E(X) = \frac{dm_X(t)}{dt}\Big|_{t=0}}$

MGF's can
(1) Find moments
(2) characterize dist

→ If we can find the MGF of a transformation of some RV's — and can recognize its form — then we know the dist. of the transformation

→ especially useful for linear transformations

Let $Y_1, \ldots, Y_n$ be ind. RVs w/ MGF's $m_{Y_i}(t)$, $i = 1, \ldots, n$. If $X = a_1Y_1 + \ldots + a_nY_n$, then

$$\boxed{m_X(t) = \prod_{i=1}^{n} m_{Y_i}(a_i t)}$$

NOTES: (·) sum of ind. Poissons is Poisson

(·) Linear transformations of Normal RV's are normal

method of Dist. Functions:

Let $X = |Y|$. since $f_X(x) = \frac{d}{dx}F_X(x)$

$F_X(x) = P(X \leq x)$ $\quad = \frac{d}{dx}[F_Y(x) - F_Y(-x)]$

$= P(|Y| \leq x)$

$= P(-x \leq Y \leq x)$ $\quad = f_Y(x) - f_Y(-x)(-1)$

$= F_Y(x) - F_Y(-x)$ $\quad = f_Y(x) + f_Y(-x)$

Fundamental theorem of stat (special case):

$\mu_1 = \mu_2 = \mu \Rightarrow \bar{Y} = \frac{1}{n}\sum X_i = \bar{X}$

$\sigma_1^2 = \sigma_2^2 = \sigma^2 \Rightarrow \boxed{\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)}$

$a_1 = \cdots = \frac{1}{n}$ → average of independent normals are normal

method of Dist. functions (cont):

Let $X = h(Y) = y^2$ $\quad$ to find CDF of $X = h(Y)$:

$P(X \leq 6) = P\{h(Y) \leq 6\}$ $\quad P(X \leq x) = \int_{-\sqrt{x}}^{\sqrt{x}} f(y)\,dy,\ x \geq 0$

$= P(Y^2 \leq 6)$

$= P(-\sqrt{6} \leq Y \leq \sqrt{6})$

$= \int_{-\sqrt{6}}^{\sqrt{6}} f(y)\,dy$

→ implementing the **MGF method** for determining the dist. of linear combos of ind. RV's

MGF EX: $X \sim Pois(\lambda)$

$m_X(t) = E(e^{tX}) = \sum_{x=0}^{\infty} e^{tx} e^{-\lambda}\frac{\lambda^x}{x!}$

$= e^{-\lambda}\sum_{x=0}^{\infty}\frac{(\lambda e^t)^x}{x!} = e^{-\lambda}e^{\lambda e^t}$

$= e^{\lambda(e^t-1)}$

$\frac{dm_X(t)}{dt} = e^{\lambda(e^t-1)} \cdot \lambda e^t \Big|_0 = \lambda = E(X)$.

Linear transformations of Normal are normal

$Y_1, \ldots, Y_n$ ind. RV's s.t. $Y_i \sim N(\mu_i, \sigma_i^2)$, $i = 1, \ldots, n$.

suppose $X = \sum_{i=1}^{n} a_iY_i$; MGF of Y_i is

$m_{Y_i}(t) = E(e^{tY_i})$

$= e^{\mu_i t + \sigma_i^2 t^2/2}$

$\Rightarrow m_{Y_i}(a_i t) = e^{a_i\mu_i t + a_i^2\sigma_i^2 t^2/2}$

$\Rightarrow m_X(t) = \prod_{i=1}^{n} m_{Y_i}(a_i t) = e^{(\sum a_i\mu_i)t + (\sum a_i^2\sigma_i^2)t^2/2}$

→ has form of normal RV w/ mean $\sum a_i\mu_i$ and var $\sum a_i^2\sigma_i^2$

$\Rightarrow X \sim N(\sum a_i\mu_i, \sum a_i^2\sigma_i^2)$

General Strat. to find Prob. Dist. of RV Function

suppose $Y \sim unif(0,1)$. Let $X = h(Y) = -\log Y$

$F_X(x) = P(X \leq x)$ $\Rightarrow f_X(x) = \frac{d}{dx}F_X(x)$

$= P(X \leq x)$ $\quad = \frac{d}{dx}(1 - e^{-x})$

$= P(-\log Y \leq x)$ $\quad = e^{-x},\ x \geq 0$

$= P(Y \geq -e^{x})$

$= 1 - F_Y(e^{-x})$ $\quad$ (always works — just not always most efficient)

$= 1 - e^{-x}$

$\boxed{m_X(t) = m_1(a_1t)\,m_2(a_2t)}$

Prob. Dist. Transf. (Discrete)

$P_Y(y) = \begin{cases} 1/6, & y = 1, \ldots, 6 \\ 0, & \text{otherwise} \end{cases}$

Let $X = (Y-4)^2$

Q: PMF of X

x	$P_X(x)$
0	$1/6 = P(Y=4)$
1	$2/6 = P(Y=3 \text{ or } Y=5)$
4	$2/6$
9	$1/6 = P(Y=1)$

Bivariate transformation:

$Y_1 \sim EXP(1)$, $Y_2 \sim EXP(1)$ — independent

$f_{Y_1,Y_2}(y_1,y_2) = f_1(y_1)f_2(y_2) = e^{-y_1}e^{-y_2}$

Let $X_1 = Y_1 + Y_2$, $X_2 = Y_2$ $\quad$ Q: find joint PDF for X

Inverse transf.: $Y_1 = X_1 - X_2$, $Y_2 = X_2$ | **note**: $\{(x_1,x_2): x_2 \leq y_1\}$

$f_{X_1,X_2}(x_1,x_2) = f_{Y_1,Y_2}(x_1 - x_2, x_2)J(x_1,x_2)$

$= e^{-(x_1-x_2)}e^{-x_2}\left|\begin{smallmatrix}1 & -1 \\ 0 & 1\end{smallmatrix}\right| \Rightarrow e^{-x_1}$

u-sub

$\int (x+1)^2\,dx$

$u = x+1$

$\frac{du}{dx} = 1 \Rightarrow du = dx$

$\int u^2\,du$

$= \frac{u^3}{3} + C$

$= \frac{(x+1)^3}{3} + C$

Table of Common Distributions

taken from *Statistical Inference* by Casella and Berger

Discrete Distrbutions

distribution	pmf	mean	variance	mgf/moment
Bernoulli(p)	$p^x(1-p)^{1-x};\ x=0,1;\ p\in(0,1)$	p	$p(1-p)$	$(1-p)+pe^t$
Beta-binomial(n,α,β)	$\binom{n}{x}\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\frac{\Gamma(x+\alpha)\Gamma(n-x+\beta)}{\Gamma(\alpha+\beta+n)}$	$\frac{n\alpha}{\alpha+\beta}$	$\frac{n\alpha\beta}{(\alpha+\beta)^2}$	
Notes: If $X\vert P$ is binomial (n,P) and P is beta(α,β), then X is beta-binomial(n,α,β).				
Binomial(n,p)	$\binom{n}{x}p^x(1-p)^{n-x};\ x=1,\ldots,n$	np	$np(1-p)$	$[(1-p)+pe^t]^n$
Discrete Uniform(N)	$\frac{1}{N};\ x=1,\ldots,N$	$\frac{N+1}{2}$	$\frac{(N+1)(N-1)}{12}$	$\frac{1}{N}\sum_{i=1}^{N}e^{it}$
Geometric(p)	$p(1-p)^{x-1};\ p\in(0,1)$	$\frac{1}{p}$	$\frac{1-p}{p^2}$	$\frac{pe^t}{1-(1-p)e^t}$
Note: $Y=X-1$ is negative binomial($1,p$). The distribution is *memoryless*: $P(X>s\vert X>t)=P(X>s-t)$.				
Hypergeometric(N,M,K)	$\frac{\binom{M}{x}\binom{N-M}{K-x}}{\binom{N}{K}};\ x=1,\ldots,K$ $M-(N-K)\le x\le M;\ N,M,K>0$	$\frac{KM}{N}$	$\frac{KM}{N}\frac{(N-M)(N-k)}{N(N-1)}$	?
Negative Binomial(r,p)	$\binom{r+x-1}{x}p^r(1-p)^x;\ p\in(0,1)$ $\binom{y-1}{r-1}p^r(1-p)^{y-r};\ Y=X+r$	$\frac{r(1-p)}{p}$	$\frac{r(1-p)}{p^2}$	$\left(\frac{p}{1-(1-p)e^t}\right)^r$
Poisson(λ)	$\frac{e^{-\lambda}\lambda^x}{x!};\ \lambda\ge 0$	λ	λ	$e^{\lambda(e^t-1)}$
Notes: If Y is gamma(α,β), X is Poisson($\frac{x}{\beta}$), and α is an integer, then $P(X\ge\alpha)=P(Y\le y)$.				

Continuous Distributions

distribution	pdf	mean	variance	mgf/moment
Beta(α, β)	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1};\ x \in (0,1),\ \alpha, \beta > 0$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	$1+\sum_{k=1}^{\infty}\left(\prod_{r=0}^{k-1}\frac{\alpha+r}{\alpha+\beta+r}\right)\frac{t^k}{k!}$
Cauchy(θ, σ)	$\frac{1}{\pi\sigma}\frac{1}{1+(\frac{x-\theta}{\sigma})^2};\ \sigma > 0$	does not exist	does not exist	does not exist
Notes: Special case of Students's t with 1 degree of freedom. Also, if X, Y are iid $N(0,1)$, $\frac{X}{Y}$ is Cauchy				
χ_p^2	$\frac{1}{\Gamma(\frac{p}{2})2^{\frac{p}{2}}} x^{\frac{p}{2}-1} e^{-\frac{x}{2}};\ x > 0,\ p \in N$	p	$2p$	$\left(\frac{1}{1-2t}\right)^{\frac{p}{2}},\ t < \frac{1}{2}$
Notes: Gamma$(\frac{p}{2}, 2)$.				
Double Exponential(μ, σ)	$\frac{1}{2\sigma} e^{-\frac{\lvert x-\mu\rvert}{\sigma}};\ \sigma > 0$	μ	$2\sigma^2$	$\frac{e^{\mu t}}{1-(\sigma t)^2}$
Exponential(θ)	$\frac{1}{\theta} e^{-\frac{x}{\theta}};\ x \geq 0,\ \theta > 0$	θ	θ^2	$\frac{1}{1-\theta t},\ t < \frac{1}{\theta}$
Notes: Gamma$(1, \theta)$. Memoryless. $Y = X^{\frac{1}{\gamma}}$ is *Weibull*. $Y = \sqrt{\frac{2X}{\beta}}$ is *Rayleigh*. $Y = \alpha - \gamma \log \frac{X}{\beta}$ is *Gumbel*.				
F_{ν_1, ν_2}	$\frac{\Gamma(\frac{\nu_1+\nu_2}{2})}{\Gamma(\frac{\nu_1}{2})\Gamma(\frac{\nu_2}{2})}\left(\frac{\nu_1}{\nu_2}\right)^{\frac{\nu_1}{2}} \frac{x^{\frac{\nu_1-2}{2}}}{\left(1+(\frac{\nu_1}{\nu_2})x\right)^{\frac{\nu_1+\nu_2}{2}}};\ x > 0$	$\frac{\nu_2}{\nu_2-2},\ \nu_2 > 2$	$2(\frac{\nu_2}{\nu_2-2})^2\frac{\nu_1+\nu_2-2}{\nu_1(\nu_2-4)},\ \nu_2 > 4$	$EX^n = \frac{\Gamma(\frac{\nu_1+2n}{2})\Gamma(\frac{\nu_2-2n}{2})}{\Gamma(\frac{\nu_1}{2})\Gamma(\frac{\nu_2}{2})}\left(\frac{\nu_2}{\nu_1}\right)^n,\ n < \frac{\nu_2}{2}$
Notes: $F_{\nu_1,\nu_2} = \frac{\chi^2_{\nu_1}/\nu_1}{\chi^2_{\nu_2}/\nu_2}$, where the χ^2s are independent. $F_{1,\nu} = t_\nu^2$.				
Gamma(α, β)	$\frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-\frac{x}{\beta}};\ x > 0,\ \alpha, \beta > 0$	$\alpha\beta$	$\alpha\beta^2$	$\left(\frac{1}{1-\beta t}\right)^\alpha,\ t < \frac{1}{\beta}$
Notes: Some special cases are exponential $(\alpha = 1)$ and χ^2 $(\alpha = \frac{p}{2}, \beta = 2)$. If $\alpha = \frac{2}{3}$, $Y = \sqrt{\frac{X}{\beta}}$ is *Maxwell*. $Y = \frac{1}{X}$ is *inverted gamma*.				
Logistic(μ, β)	$\frac{1}{\beta}\frac{e^{-\frac{x-\mu}{\beta}}}{\left[1+e^{-\frac{x-\mu}{\beta}}\right]^2};\ \beta > 0$	μ	$\frac{\pi^2\beta^2}{3}$	$e^{\mu t}\Gamma(1+\beta t),\ \lvert t\rvert < \frac{1}{\beta}$
Notes: The cdf is $F(x \mid \mu, \beta) = \frac{1}{1+e^{-\frac{x-\mu}{\beta}}}$.				
Lognormal(μ, σ^2)	$\frac{1}{\sqrt{2\pi}\sigma}\frac{1}{x} e^{-\frac{(\log x-\mu)^2}{2\sigma^2}};\ x > 0, \sigma > 0$	$e^{\mu+\frac{\sigma^2}{2}}$	$e^{2(\mu+\sigma^2)} - e^{2\mu+\sigma^2}$	$EX^n = e^{n\mu+\frac{n^2\sigma^2}{2}}$
Normal(μ, σ^2)	$\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}};\ \sigma > 0$	μ	σ^2	$e^{\mu t+\frac{\sigma^2 t^2}{2}}$
Pareto(α, β)	$\frac{\beta\alpha^\beta}{x^{\beta+1}};\ x > \alpha,\ \alpha, \beta > 0$	$\frac{\beta\alpha}{\beta-1},\ \beta > 1$	$\frac{\beta\alpha^2}{(\beta-1)^2(\beta-2)},\ \beta > 2$	does not exist
t_ν	$\frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})}\frac{1}{\sqrt{\nu\pi}}\frac{1}{(1+\frac{x^2}{\nu})^{\frac{\nu+1}{2}}}$	$0,\ \nu > 1$	$\frac{\nu}{\nu-2}, \nu > 2$	$EX^n = \frac{\Gamma(\frac{\nu+1}{2})\Gamma(\nu-\frac{n}{2})}{\sqrt{\pi}\Gamma(\frac{\nu}{2})}\nu^{\frac{n}{2}},\ n$ even
Notes: $t_\nu^2 = F_{1,\nu}$.				
Uniform(a, b)	$\frac{1}{b-a},\ a \leq x \leq b$	$\frac{b+a}{2}$	$\frac{(b-a)^2}{12}$	$\frac{e^{bt}-e^{at}}{(b-a)t}$
Notes: If $a = 0,\ b = 1$, this is special case of beta $(\alpha = \beta = 1)$.				
Weibull(γ, β)	$\frac{\gamma}{\beta} x^{\gamma-1} e^{-\frac{x^\gamma}{\beta}};\ x > 0,\ \gamma, \beta > 0$	$\beta^{\frac{1}{\gamma}}\Gamma(1+\frac{1}{\gamma})$	$\beta^{\frac{2}{\gamma}}\left[\Gamma(1+\frac{2}{\gamma}) - \Gamma^2(1+\frac{1}{\gamma})\right]$	$EX^n = \beta^{\frac{n}{\gamma}}\Gamma(1+\frac{n}{\gamma})$
Notes: The mgf only exists for $\gamma \geq 1$.				

Midterm II Formula Sheet

If X, Y two random variables, a, b two scalars, then

$$Cov(X,Y) = E(XY) - E(X)E(Y), \quad Cor(X,Y) = \frac{Cov(X,Y)}{\sqrt{Var(X)}\sqrt{Var(Y)}},$$
$$Var(aX + bY) = a^2 Var(X) + b^2 Var(Y) + 2ab Cov(X,Y).$$

Transformation theorem: if $Y = g(X)$, then $f_Y(y) = f_X\{g^{-1}(y)\}|\frac{d}{dy}g^{-1}(y)|$, where $g^{-1}(y) = x$.
if X a random variable with $E(X) = \mu$, then $Var(X) = E(X^2) - \mu^2$.
Markov inequality: if X is a non-negative random variable and $E(X)$ exists, then $P(X \geq t) \leq E(X)/t$.
Moment generating function of a random variable X is $M(t) = E(e^{tX})$ for $t \in \mathbb{R}$. Also, $E(X) = M'(0)$.

If $Z \sim N(0,1)$ and $U \sim \chi^2_m$ are independent, then $\frac{Z}{\sqrt{U/m}} \sim t_m$.
If $U \sim \chi^2_m$ and $V \sim \chi^2_n$ are independent, then $\frac{U/m}{V/n} \sim F_{m,n}$.

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2.$$

If $X_i \overset{iid}{\sim} N(\mu, \sigma^2)$, then $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$.

Mean squared error: $MSE(\hat{\theta}) = E\{(\hat{\theta} - \theta)^2\} = Var(\hat{\theta}) + \{Bias(\hat{\theta})\}^2$.
Error of estimation: $\epsilon = |\hat{\theta} - \theta|$.

$z_{\alpha/2}$ is a critical value of $N(0,1)$ means that, for $Z \sim N(0,1)$,

$$P(Z < -z_{\alpha/2}) = P(Z > z_{\alpha/2}) = \frac{\alpha}{2}, \quad P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) = 1 - \alpha.$$

$\chi^2_{\alpha/2,d}$ is a critical value of χ^2_d means that, for $U \sim \chi^2_d$, $P(U > \chi^2_{\alpha/2,d}) = \alpha/2$.
If $\hat{\theta}$ is unbiased for θ, solve $B = z_{\alpha/2}\sigma_{\hat{\theta}}$ to find the sample size to estimate θ within B.
A $100(1-\alpha)\%$ CI for $\mu_1 - \mu_2$, under some assumptions, is

$$(\bar{X}_{n_1} - \bar{Y}_{n_2}) \pm t_{\alpha/2, n_1+n_2-2} S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}, \text{ where } S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1 + n_2 - 2}.$$

A $100(1-\alpha)\%$ small sample CI for μ is $\bar{X}_n \pm t_{\alpha/2,n-1}S/\sqrt{n}$ under some assumptions.
A $100(1-\alpha)\%$ CI for σ^2 is $\{(n-1)S^2/\chi^2_{\alpha/2,n-1}, (n-1)S^2/\chi^2_{1-\alpha/2,n-1}\}$ under some assumptions.

Relative efficiency: $\text{eff}(\hat{\theta}_1, \hat{\theta}_2) = Var(\hat{\theta}_2)/Var(\hat{\theta}_1)$.
Consistency: $\lim_{n\to\infty} P(|\hat{\theta}_n - \theta| > \epsilon) = 0$ for any positive ϵ. Also, $\lim_{n\to\infty} Var(\hat{\theta}_n) = 0$.

Likelihood $L(\theta; y_1, \ldots, y_n)$ is the joint probability/density of $Y_1, \ldots, Y_n$ evaluated at $y_1, \ldots, y_n$.
Sufficiency: U is sufficient for θ if the conditional distribution of $X_1, \ldots, X_n$ given U does not depend on θ.

Equivalently, U is sufficient for θ if $L(\theta) = g(u, \theta)h(y_1, \ldots, y_n)$, where $g(u, \theta)$ is a function of the statistic and θ, and $h(y_1, \ldots, y_n)$ does not depend on θ.
Rao-Blackwell: U sufficient, $\tilde{\theta} = E(\hat{\theta}|U)$, then $MSE(\tilde{\theta}) \leq MSE(\hat{\theta})$.

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kth moment: $\mu_k = E(X^k)$; kth sample moment: $\hat{\mu}_k = \frac{1}{n}\sum_{i=1}^n X_i^k$.
MLE $\hat{\theta}$ property: $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, I^{-1}(\theta))$ as $n \to \infty$
Fisher information: $I(\theta) = E\left\{\frac{\partial}{\partial\theta} \log\left[f(X;\theta)\right]\right\}^2 = -E\left[\frac{\partial^2}{\partial\theta^2} \log\left[f(X;\theta)\right\}\right]$

Type I error α: rejecting H_0 when H_0 is true
Type II error β: failing to reject H_0 when H_A is true
Test statistic $Z_0 = \frac{\hat{\theta}-\theta_0}{\sigma_{\hat{\theta}}}$